

BEAUFORT, NC, USA

WMO: 723037

Lat: 34.733N

Lon: 76.657W

Elev: 8

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 93765

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.4	28.6	8.7	8.6	29.3	13.1	10.7	33.0	27.8	44.8	25.6	42.8	10.3	0	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	88.3	78.5	87.0	78.0	85.7	77.5	81.3	85.5	80.4	84.5	79.6	83.7	12.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.0	155.7	83.9	79.1	151.1	83.4	78.2	146.5	82.9	44.8	85.8	43.8	84.7	42.9	83.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.6	21.0	19.1	19.3	92.5	4.5	3.7	16.0	95.2	13.4	97.3	10.8	99.4	7.5	102.1		
			16.4	82.7	3.8	1.1	13.7	83.5	11.4	84.1	9.3	84.7	6.5	85.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.9	45.5	47.7	53.6	62.3	70.3	77.6	80.4	79.6	75.2	66.3	56.3	50.7
	DBStd	14.03	9.70	8.22	8.01	6.63	5.56	3.83	3.06	3.12	4.23	7.44	7.90	8.80
	HDD50	525	204	129	56	3	0	0	0	0	0	1	29	103
	HDD65	2394	603	484	359	126	20	0	0	0	1	76	276	449
	CDD50	5593	66	66	168	371	629	828	942	916	757	508	219	123
	CDD65	1986	0	1	6	44	184	378	477	451	308	117	16	4
	CDH74	16379	0	0	8	64	801	3264	4998	4470	2161	579	34	0
	CDH80	3940	0	0	2	2	74	684	1468	1260	397	51	2	0
Wind	WSAvg	9.5	9.7	10.2	10.5	10.9	10.0	9.8	9.9	8.1	8.5	8.5	8.8	9.1
Precipitation	PrecAvg	57.8	4.9	4.2	4.1	3.2	4.8	4.7	6.9	6.8	6.1	4.2	3.7	4.1
	PrecMax	91.5	9.6	12.4	11.3	9.3	13.4	10.9	14.5	15.5	15.8	13.0	10.6	11.5
	PrecMin	35.0	1.4	1.3	0.3	0.3	1.1	0.3	1.7	1.4	0.3	0.2	0.9	1.1
	PrecStd	11.0	2.3	2.4	2.1	2.0	2.8	2.4	3.0	3.7	3.8	3.1	2.2	2.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.2	71.2	74.0	78.8	84.7	89.7	90.7	90.8	87.0	84.0	77.7	72.2
		MCWB	64.5	64.4	65.0	70.8	74.6	76.7	79.3	78.8	75.6	76.2	69.2	68.3
	2%	DB	66.1	66.5	71.0	75.7	81.7	87.0	88.3	88.1	85.2	81.3	73.8	69.7
		MCWB	62.5	61.0	63.2	68.2	73.9	76.9	79.5	78.6	75.6	74.5	68.8	66.2
	5%	DB	63.8	63.8	68.3	73.7	80.0	85.1	86.9	86.6	83.9	79.3	72.0	67.5
		MCWB	60.7	59.4	61.8	67.1	72.8	76.5	78.7	78.0	75.3	72.9	67.0	64.2
	10%	DB	61.1	61.2	66.0	72.1	78.5	83.5	85.7	85.3	82.3	77.3	69.6	65.1
		MCWB	57.9	57.2	60.8	65.9	71.5	75.9	78.2	77.5	74.9	71.3	64.8	62.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.9	66.2	67.2	73.5	77.2	81.1	82.4	82.4	80.0	78.6	73.1	69.8
		MCDB	67.6	69.9	70.9	76.1	82.2	84.9	87.2	86.5	83.5	81.0	75.8	71.3
	2%	WB	63.4	63.3	65.4	70.7	75.5	79.9	81.5	81.2	78.4	76.2	70.8	67.1
		MCDB	65.4	65.2	69.3	74.1	80.2	84.1	86.0	85.3	82.4	79.8	72.8	68.6
	5%	WB	61.2	60.5	63.7	68.8	74.3	78.6	80.6	80.2	77.6	74.5	67.8	65.0
		MCDB	63.6	63.0	67.1	72.2	78.7	83.1	84.9	84.3	81.9	78.4	70.5	66.7
	10%	WB	58.4	57.7	61.9	67.0	73.0	77.7	79.9	79.2	76.5	72.8	65.4	62.3
		MCDB	61.0	60.7	65.1	70.5	77.1	82.3	84.1	83.4	80.8	76.5	68.2	64.7
Mean Daily Temperature Range	5% DB	MDBR	16.1	16.1	15.9	14.0	12.3	10.8	10.1	10.8	11.9	14.4	16.9	15.9
		MCDBR	17.7	17.4	16.7	13.4	11.9	11.5	10.7	11.4	12.3	13.0	16.9	15.4
		MCWBR	15.0	13.7	11.2	8.4	6.5	6.3	6.1	5.9	6.2	8.1	12.0	12.8
	5% WB	MCDBR	16.3	16.1	13.9	11.5	10.0	9.8	9.2	9.5	10.0	11.1	15.2	13.9
		MCWBR	14.8	14.8	10.7	8.6	6.1	6.2	6.1	6.1	6.1	8.2	13.1	12.9
Clear-Sky Solar Irradiance	taub	0.327	0.336	0.362	0.401	0.445	0.490	0.515	0.499	0.424	0.373	0.348	0.336	
	taud	2.540	2.509	2.455	2.345	2.266	2.186	2.126	2.167	2.383	2.499	2.518	2.536	
	Ebn at Noon	280	289	289	281	269	256	248	250	266	272	269	269	
	Edh at Noon	26	30	34	39	43	47	49	47	36	30	26	25	
All-Sky Solar Radiation	RadAvg	814	1049	1391	1764	1956	2021	1905	1736	1433	1217	906	725	
	RadStd	49	101	104	124	130	111	116	102	94	122	63	53	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+306	+211

Nomenclature: See separate page